Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Code:

```
clc; clear; close all;

%% ==========================================
% FIR High-Pass Filter Design Using
% Rectangular, Hamming, and Hann Windows
%% ==========================================

% Input Parameters
fp = 1000;  % High-pass cutoff frequency (Hz) Fs = 8000;         % Sampling
frequency (Hz)
N = 50;     % Filter order
% Normalized cutoff frequency Wn = fp/(Fs/2);

%% 1. FIR High-Pass Filter using Rectangular Window b_rect = fir1(N, Wn,
'high', rectwin(N+1));

disp('=========================================');
disp('High-Pass FIR Filter (Rectangular Window)');
disp('=========================================');
disp(b_rect');
%% 2. FIR High-Pass Filter using Hamming Window b_hamm = fir1(N, Wn, 'high',
hamming(N+1));

disp('=====================================');
disp('High-Pass FIR Filter (Hamming Window)');
disp('=====================================');
disp(b_hamm');

%% 3. FIR High-Pass Filter using Hann Window
b_hann = fir1(N, Wn, 'high', hann(N+1));  % Use hanning(N+1) if required

disp('==================================');
disp('High-Pass FIR Filter (Hann Window)');
disp('==================================');
disp(b_hann');

%% Display Filter Information fprintf('\nFilter Order : %d\n', N);
fprintf('Sampling Frequency : %d Hz\n', Fs); fprintf('Cutoff Frequency
      : %d Hz\n', fp); fprintf('Normalized Cutoff : %.2f\n\n', Wn);

%% Frequency Response - Rectangular Window figure;
freqz(b_rect,1,1024,Fs);
title('High-Pass FIR using Rectangular Window');

%% Frequency Response - Hamming Window figure;
freqz(b_hamm,1,1024,Fs);
title('High-Pass FIR using Hamming Window');

%% Frequency Response - Hann Window figure;
freqz(b_hann,1,1024,Fs);
```

```
title('High-Pass FIR using Hann Window');

%% Magnitude Response Comparison [H1,f] = freqz(b_rect,1,1024,Fs); [H2,~] =
freqz(b_hamm,1,1024,Fs); [H3,~] = freqz(b_hann,1,1024,Fs);

figure; plot(f,abs(H1),'LineWidth',1.5); hold on;
plot(f,abs(H2),'LineWidth',1.5);
plot(f,abs(H3),'LineWidth',1.5); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude');
title('Comparison of FIR High-Pass Filters');
legend('Rectangular','Hamming','Hann'); xlim([0 Fs/2]);
```

Output:

Fig 1:

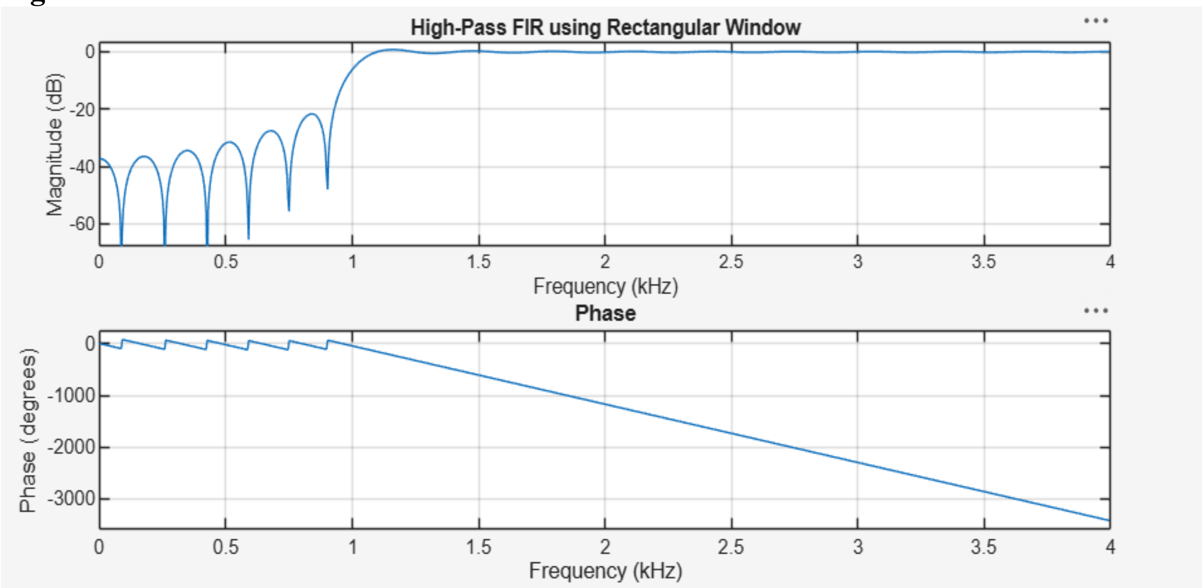


Fig 2:

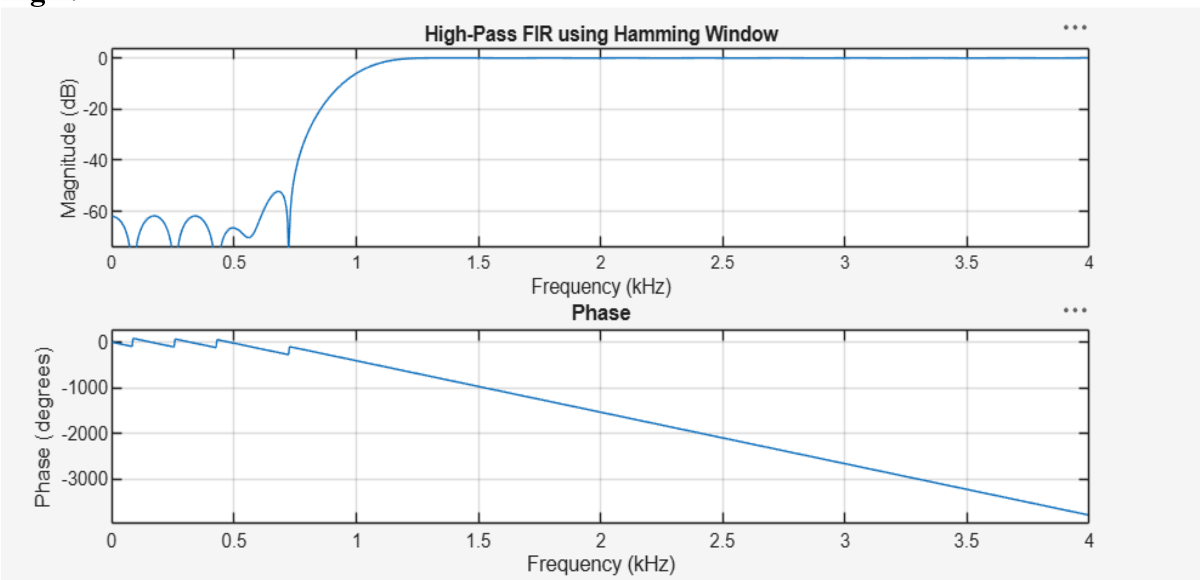


Fig 3:

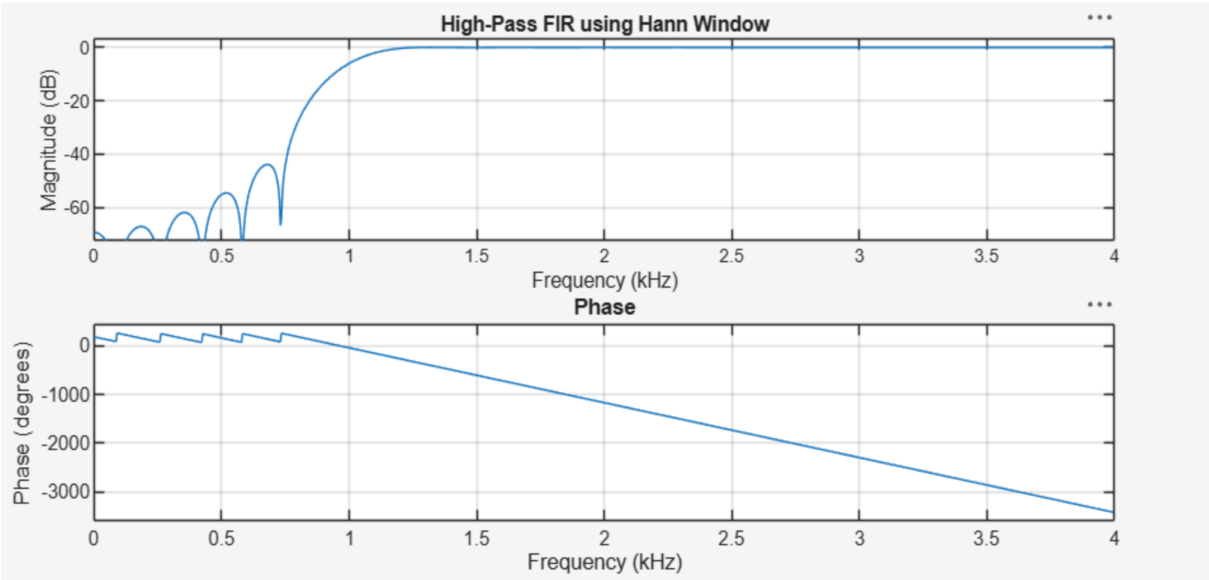
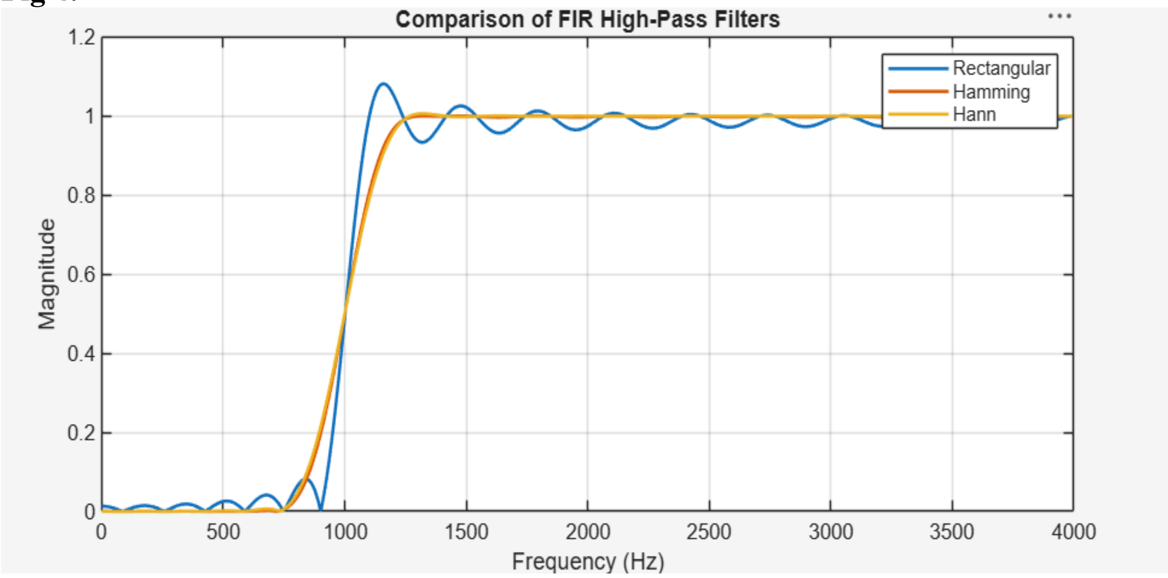

High-Pass FIR using Hann Window
Magnitude (dB)
0
-20
-40
-60
0
0.5
1
1.5
2
2.5
3
3.5
4
Frequency (kHz)
Phase
Phase (degrees)
0
-1000
-2000
-3000
0
0.5
1
1.5
2
2.5
3
3.5
4
Frequency (kHz)

Fig 4:


Comparison of FIR High-Pass Filters
Magnitude
1.2
1
0.8
0.6
0.4
0.2
0
0
500
1000
1500
2000
2500
3000
3500
4000
Frequency (Hz)
Rectangular
Hamming
Hann

Expt 5	**Design of High Pass Filter using Windowing Techniques**
Date:	**5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques**

Code:

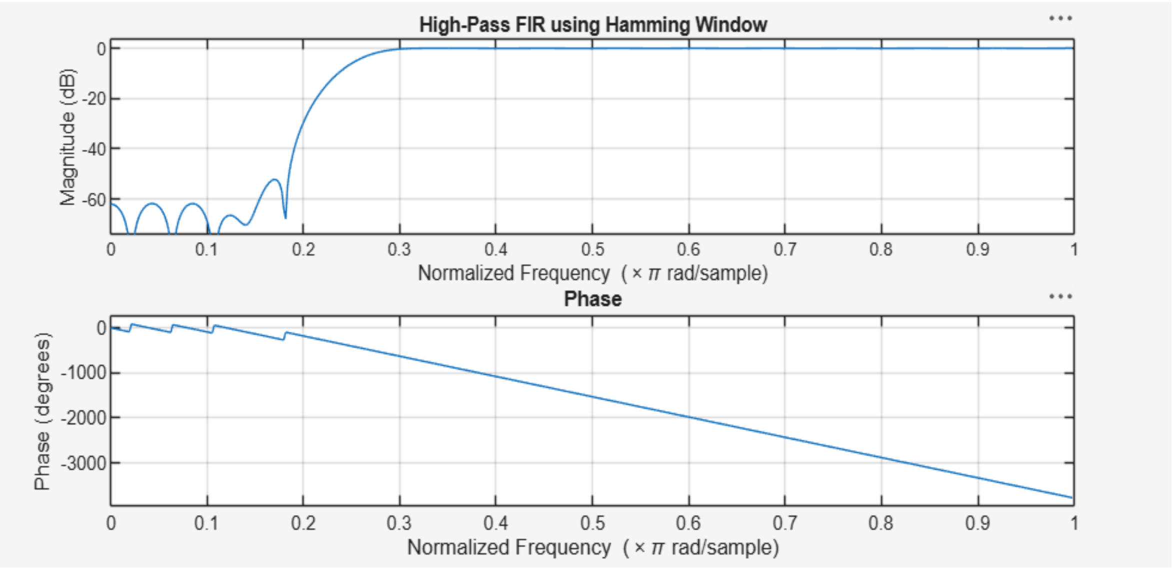
```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000;  % High-pass cutoff frequency (Hz) fs = 8000;    %
Sampling frequency (Hz)
N = 50;    % Filter order
wc = fp/(fs/2);  % Normalized cutoff (0–1)
%% --- FIR High-Pass using Hamming Window ---b_hamm = fir1(N, wc,
'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');

%% --- Frequency Response Plot ---figure;
freqz(b_hamm,1);
title('High-Pass FIR using Hamming Window');
```

Output:

Fig 1:



Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

Code:

```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000;   % High-pass cutoff frequency (Hz) fs = 8000;   % Sampling
frequency (Hz)
N = 50;     % Filter order
% Normalized cutoff frequency wc = fp/(fs/2);
%% --- FIR High-Pass Filter using Hanning Window ---b_hann = fir1(N, wc,
'high', hanning(N+1));
% Display filter coefficients
disp('--- High-Pass Coefficients (Hanning Window) ---'); disp(b_hann');
%% --- Frequency Response ---figure;
freqz(b_hann,1);
title('High-Pass FIR using Hanning Window');
```

Output:

Fig 1: